Shambhavi Mishra

+91 8542094746 | shambhavi16204@gmail.com | linkedin.com/in/shambhavi-mishra | github.com/Diya-16204

Professional Summary

CS undergrad (2027) at IIIT Guwahati with experience building full-stack and ML systems, two research internships, and **500+** DSA problems solved. Interested in scalable backend and distributed systems.

Education

Indian Institute of Information Technology, Guwahati — BTech, CSE | CPI: **8.19** — Aug **2023** – Jun **2027**
Dayawati Modi Academy, Rampur — Class XII (CBSE) | **97.2%** — **2022**
Shri Gulab Rai Montessori School, Bareilly — Class X (CBSE) | **97.4%** — **2020**

Technical Skills

Languages & Databases: Java, Python, C, JavaScript | MySQL, MongoDB
Web Development: React.js, HTML5, CSS3, Tailwind CSS, Bootstrap, Node.js, Express.js
Machine Learning: NumPy, Pandas, Matplotlib, TensorFlow, Keras
Tools & Coursework: Git, GitHub, Linux, Windows | DSA, OOPs, DBMS, OS , Software Engineering

Projects

FinPilot — Personal Finance Management System [GitHub] **2025**

- Built a **MERN** finance platform with **12**+ REST APIs and **JWT** auth, serving **50**+ users; added indexed MongoDB queries and pagination to keep dashboard loads fast.
- Secured sensitive financial data with JWT-based authentication and strict authorization protocols.
- Crafted interactive user dashboards on top of the APIs, decreasing manual expense tracking effort by **70%**.

Automated Digital Forensics Evidence Collection Tool [GitHub] **2026**

- Built an evidence collection pipeline with **RBAC** and **SHA-256** hash verification that detected all tampering in test cases; cut manual data handling by **40%**.
- Architected a layered system design using **UML diagrams**, maintaining **90%+** test coverage.

GuardianCell — Multimodal Clinical Support System [GitHub] **2026 Ongoing**

- Engineered a multimodal ML architecture processing tabular data and imaging for lesion classification and subtyping.
- Fine-tuned **DenseNet121** and **EfficientNetB0** backbones, improving accuracy by **4–5%** over baseline models.
- Constructed feature-fusion and preprocessing pipelines to handle missing modalities and mitigate class imbalance.

Experience

IIT BHU [Certificate] **June 2026 – July 2026**
Project Intern — Multimodal Healthcare AI *Varanasi, India*

- Designed cascading diagnostic frameworks combining clinical records with triple-modality imaging (mammography, ultrasound, MRI) to streamline breast cancer workflows.
- Co-designed decision support protocols to non-invasively project **Oncotype DX** recurrence risk and guide post-treatment chemotherapy planning.
- Validated prognostic pipelines against clinical benchmark datasets to model long-term post-treatment survival probabilities.

IIT Guwahati [Certificate] **May 2026 – June 2026**
Research Intern — Vision & Security *Guwahati, India*

- Implemented a universal adversarial patch attack (based on **SlowFormer, CVPR 2024**) that increased inference compute on dynamic vision transformers while preserving top-1 accuracy.
- Evaluated **Adversarial Efficiency Degradation Attacks** on input-dependent dynamic networks, including ATS, A-ViT, and AdaViT variants.
- Evaluated defenses against compute-exhaustion attacks, analyzing **Adversarial Training** and token-distribution regularization to mitigate FLOP expansion.

Honours & Achievements

- **Problem Solving:** Solved **500+** Data Structures & Algorithms problems across various coding platforms.
- **Selected Scholar — Amazon ML Summer School (MLSS) '26:** Selected for cohort of 3k from **1.3L+** applicants (Top **2%**).
- **Finalist — Myntra WeForShe Hackathon:** Ranked in **Top 100** out of **55,000+** competitors (Top **0.18%** nationwide).
- **Global Cohort Scholar — SheFi 2025:** Selected from **3,200+** worldwide applicants for an intensive technical cohort.
- **Contributor — GirlScript Summer of Code 2025:** Authored and merged core code patches across production repositories.

Position of Responsibility

USG IT Technical Head, MUN — IIIT Guwahati **Jan 2025 – July 2025**

- Led technical operations and planning for institute-level MUN events hosting over **300**+ participants.
- Coordinated workloads across **10**+ internal teams and digitized operations workflows to cut manual overheads by **50%**.

Programming Club Coordinator, Technical Board — IIIT Guwahati **Aug 2024 – Jan 2025**

- Organized **6**+ programming contests and peer workshops for the institute's student community.
- Mentored **200+** peers and collaborated with student boards to boost technical participation by **40%**.